

Predicting Passenger Fare Strategies with Incomplete Data

Objective:

You are a **Data Analyst** at a transport tech company studying fare optimization and passenger demographics. Your dataset has missing values in crucial columns like **Age**, **Fare**, and **Cabin**. Your job is to:

- Detect and handle missing data properly.
- Understand how different methods of imputation affect business decisions.
- Prepare the data for downstream modeling (e.g., predicting fare range or passenger segmentation).

Dataset: [bus fare dataset.csv](#)

Colab Notebook [link](#)

Here is the description of each column in your dataset:

- **PassengerId**: Unique identifier for each passenger
- **Pclass**: Passenger class (1 = 1st, 2 = 2nd, 3 = 3rd) – often used as a proxy for socioeconomic status
- **Name**: Name of the passenger
- **Sex**: Gender of the passenger (male/female)
- **Age**: Age of the passenger in years
- **SibSp**: Number of siblings or spouses aboard the ship

- **Parch:** Number of parents or children aboard the ship
 - **Ticket:** Ticket number (can indicate group or booking information)
 - **Fare:** Fare paid for the ticket (in currency units)
 - **Cabin:** Cabin number (many missing values, could indicate luxury or location on ship)
 - **Embarked:** Port of Embarkation
 - **C** = Cherbourg
 - **Q** = Queenstown
 - **S** = Southampton
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Business Value & Insights

1. **Revenue Accuracy:** Cleaning the Fare column avoids underreporting or misbilling.
2. **Customer Segmentation:** Filling Age allows clustering passengers by age group for targeted services.
3. **Pricing Strategy:** Fare imputation helps in dynamic pricing models (e.g., Fare vs Age, Fare vs Class).
4. **Marketing:** Understanding who pays more and why can shape discounts or loyalty campaigns.